

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 06

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) (1) • They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone (1) • The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface (1)	3
01.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Provides carbon, oxygen, hydrogen and nitrogen (as N₂, O₂, CO₂, H₂O) for biological molecules (1) • Delays the escape of infrared energy, keeping the surface warm (1) • Distributes heat and water vapour by winds / ocean currents (1) • Provides atmospheric pressure so liquid water can exist (1) • Deflects damaging radiation of the solar wind in the upper atmosphere (1)	2
01.3	<i>1 mark.</i> • Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers (1)	1

Question	Marking guidance	Marks
01.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Troposphere: enhanced greenhouse effect - anthropogenic CO₂ (fossil fuels, deforestation), methane (rice, livestock, landfill), N₂O, CFCs, tropospheric ozone; more infrared absorbed and re-emitted; aerosols and smoke increase albedo / scatter light; land-use change alters surface albedo and evapotranspiration • Stratosphere: CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O₂ / O₃ equilibrium; less UV absorbed in the stratosphere, more reaches the surface; stratospheric cooling • Consequences: changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions; increased UV damage to organisms • Difficulties of prediction: linked processes where one change alters others; feedback (positive and negative) and tipping points; time lags (ocean heat capacity, long residence times of CFCs and CO₂); natural variability (solar output, volcanoes, Milankovitch cycles); incomplete understanding of atmospheric processes; limited historical data so proxies needed; models with different assumptions • Evaluation: some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain; precautionary principle • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	
02.1	<i>1 mark for each point.</i> • Many different organic substances may be present, each with a different potential to cause deoxygenation, so it is difficult to measure or forecast them individually (1) • BOD measures the effect (amount of oxygen used by microbes decomposing the organic matter) under standardised conditions (eg 5 days, 20 °C, dark) (1) • It gives a single measure directly related to the risk of deoxygenation (1)	3
02.2	<i>1 mark for each point.</i> • E. coli is a common gut bacterium so its presence confirms faecal contamination and its number indicates how serious the contamination is (1) • Pathogens are not always present so their absence would not prove the water is uncontaminated (1)	2

Question	Marking guidance	Marks
02.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Different species have different sensitivities to pollution (1) • Easy to identify (1) • Easy to find / normally present / usually common (1) • Widely distributed (1)	3
02.4	<i>1 mark for each point.</i> • Advantage: assesses the effect of pollution over a period of time (recent pollution events are detected even if the pollutant has gone); no expensive equipment needed (1) • Disadvantage: identification needs skill; does not identify which pollutant is present so further chemical tests are required; species may be absent for other reasons (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9.	
03.1	<i>1 mark for each point.</i> • Readings drift so they may not be accurate; calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions (1) • pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly (1) • If a meter cannot be adjusted, record the percentage error and correct the results (1)	3
03.2	<i>1 mark for each point.</i> • Same height above the ground (1) • No obstacles to air flow; impeller type facing into the wind (1) • Axis of rotation horizontal (impeller) or vertical (cup); same time period / averaging (1)	3
03.3	<i>1 mark for each point.</i> • Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling (1) • Evaporation cools the wet bulb more when humidity is low; the temperature difference is compared with a table to find humidity (1)	2
03.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Test strips with reagent bands compared with a colour chart (semi-quantitative) (1) • Electronic colourimeter: reagent added and colour intensity measured at a specific wavelength (1) • Ion-selective electrode measuring electrode potential (other ions can interfere) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
04.1	<i>1 mark for the correct answer.</i> • $4.2 / 0.7 = 6$ (1)	1
04.2	<i>1 mark for each point.</i> • Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low (1) • In winter days are very short / the Sun does not rise, so light levels are extremely low (1)	2

Question	Marking guidance	Marks
04.3	1 mark for each point. • Reflectors can only focus parallel rays of direct sunlight onto the absorber (1) • Clouds, smoke and dust scatter the light so it cannot be focused and intensified (1)	2
04.4	1 mark. • Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further) (1)	1
04.5	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Problems: intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • Photothermal: evacuated tubes reduce conduction / convection losses; thermal stores (water, sand, concrete, molten salt) allow later use; passive solar architecture, brise soleil, light tubes; heat pumps deliver up to four times the energy used • Photovoltaic developments: multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • Integration: grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • Evaluation: strong in sunny deserts / low latitudes, weaker in high latitudes in winter; cost falling; manufacturing impacts; no single technology solves all problems; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
05.1	Correct answer only. • Its total annual discharge (1)	1

Question	Marking guidance	Marks
05.2	<i>1 mark for each point.</i> - Public supplies cannot match consumer demand / the water is often cleaner than rivers (1) - It also reduces urban flooding after heavy rain; surplus can be used for aquifer recharge (1)	2
05.3	<i>1 mark for each point.</i> - The reservoir is created in the estuary so no farmland / settlements need to be flooded (1) - But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
06.1	<i>1 mark for the correct answer.</i> $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$ (1)	1
06.2	<i>1 mark for each point.</i> - Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ (1) - Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter (1)	2
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Low / zero tillage (1) - Adding manure, compost or mulch (1) - Growing cover crops / leaving crop residues (1) - Converting to long-term grass or tree crops in rotation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
07.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed; hydrodynamics for ships (2) - Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption (2) - Conventional friction brakes convert kinetic energy to waste heat; regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again (2)	6
07.2	<i>1 mark for each point.</i> - Pneumatic tyres deform as they rotate; the continual change in shape and air movement inside generates frictional heat (more if under-inflated) (1) - Steel wheels do not deform so rolling resistance is very low (1)	2
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel (1) - Better temperature control (thermostatic fan) (1) - Electronic ignition control / regular servicing to keep spark timing correct (1) - Turbochargers supplying more oxygen (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	
08.1	<i>1 mark for each point.</i> • The void fills with water naturally; wetlands are uncommon / valuable habitats in most regions (1) • The new habitat may have greater wildlife value than the farmland lost; low cost compared with filling and levelling (1)	2
08.2	<i>1 mark for each point.</i> • Housing: ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) (1) • Farmland: landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) (1)	2
08.3	<i>1 mark.</i> • Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost; transplanted plants often fail (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	

Question	Marking guidance	Marks
09.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries • Evidence of limited success: global CO₂ concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies • Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater	9
09.2	<i>1 mark.</i> • The total greenhouse gas / CO₂ emissions caused by an individual, organisation, product or country (often expressed per capita per year) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Features: renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts • Fossil fuels: high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO₂), politically concentrated (Middle East oil) • Nuclear: very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition • Renewables: low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs • Sustainability concepts: impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion) • Economic and political factors: subsidies, guaranteed prices, planning, external costs not included in price, energy security • Judgement: a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable; transition must start before depletion	9
10.2	<i>1 mark.</i> • Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.	

Question	Marking guidance	Marks
11.1	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25

Question	Marking guidance	Marks
11.2	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25